The second memory coefficient and a quartic singlet
First returns, descendants and a complete two-point kernel

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Abstract

The second energy-centered memory coefficient admits an exact expression in fourth fermion derivatives, lower descendants and a nonlinear first-return term. Its fourth Gaussian-chaos component contains a direction independent of four descendant directions. A complete finite endpoint classification cancels all contractions with fewer than four bridges and identifies the resulting two-point measure with the established $r = q^4$ Mehler probe kernel. Separate endpoint certificates give vanishing two-energy cross measures with the quadratic and cubic singlets. The remaining quadratic chaos of the raw second memory coefficient is kept as an explicit open task.

1 Setting and the exact memory expansion

Use the standard CAR complex Gaussian p-body Hamiltonian $H = H_s$, including overlapping index sets, with variance $w = 4^p(J_s^2 N/p^2)(p!/N^p)^2$. Brackets mean the annealed normalized trace at $\beta = \mu = 0$, and $\hbar = 1$. First take $p^2/N \to \lambda > 0$, $p, N \to \infty$, with $q = e^{-4\lambda}$ and $\sigma^2 = J_s^2/\lambda$. Time is fixed in string units; small λ is a subsequent limit.

In charge $m > 0$, retain $h = H_m$, $K = 1_N \otimes H_{m-1}$ and $V\psi = m^{-1/2}\sum_i |i\rangle c_i \psi$. Write $Q = 1 - VV^\dagger$, $\mathcal{R} = QKV$, $K_\perp = QKQ$ and

$$\widehat{\mathcal{M}}_m(s) = e^{-ihs/2}\mathcal{R}^\dagger e^{iK_\perp s}\mathcal{R}e^{-ihs/2} = G + isZ_1 - \tfrac{1}{2}s^2 Z_2 + O(s^3). \tag{1}$$

Let $B_n = V^\dagger K^n V$ and $b_n = B_n - \sum_{j=1}^{n-1} b_j B_{n-j}$. The ordered first-return recursion, equivalent to an interval-partition moment relation [1], gives

$$b_4 = B_4 - B_1B_3 - B_2^2 - B_3B_1 + B_1^2B_2 + B_1B_2B_1 + B_2B_1^2 - B_1^4, \tag{2}$$

$$Z_2 = b_4 - 2\mathcal{J}_h b_3 + \mathcal{J}_h^2 G, \qquad \mathcal{J}_h X = (hX + Xh)/2. \tag{3}$$

Using this algebra does not assume Boolean independence of the microscopic Hamiltonian or its neighbouring charge blocks.

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2 Fermion derivatives and the first-return term

Define $\mathcal{L} = [H, \cdot]$, $B^{\mathrm{f}}_{ab} = \sum_i (\mathcal{L}^a c_i)^\dagger \mathcal{L}^b c_i$, $A = B^{\mathrm{f}}_{11}$, $M_n = B^{\mathrm{f}}_{0n}$ and $T_n = (M_n + M_n^\dagger)/2$. The product rule is

$$\mathcal{L} B^{\mathrm{f}}_{ab} = -B^{\mathrm{f}}_{a+1,b} + B^{\mathrm{f}}_{a,b+1}. \tag{4}$$

Starting from $M_2 = A$, it gives

$$M_4 = B^{\mathrm{f}}_{31} - 3B^{\mathrm{f}}_{22} + 3B^{\mathrm{f}}_{13}, \tag{5}$$

$$T_4 = B^{\mathrm{f}}_{22} + 2\mathcal{L}^2 A, \qquad M_4 - T_4 = 2\mathcal{L} T_3. \tag{6}$$

The fourth centered bilocal coefficient is

$$U_4 = T_4 - \tfrac{3}{2}\mathcal{L}^2 A = B^{\mathrm{f}}_{22} + \tfrac{1}{2}\mathcal{L}^2 A. \tag{7}$$

Substituting the compressed moments into the first-return recursion and setting $r = p/m$, one obtains the finite-N identity

$$\boxed{m Z_2 = B^{\mathrm{f}}_{22,m} + \frac{(1+r)^2}{4}\mathcal{L}_h^2 A_m + 2r\mathcal{J}_h T_{3,m} + 3r^2 \mathcal{J}_h^2 A_m - \frac{A_m^2}{m} - m r^4 h^4}, \tag{8}$$

where $\mathcal{L}_h = [h, \cdot]$. All terms are Hermitian. In particular, $-A_m^2/m$ cannot be omitted from the finite-system formula.

The effect of the return and drift terms can be isolated exactly. Use the previous conserved pieces $K_A = a_m + c_m(h^2 - C_m)$ and P_m, and set $Y = A_m - K_A$, $U = T_{3,m} - P_m$. Define

$$D_{2,m} = 2r\mathcal{J}_h P_m + 3r^2 \mathcal{J}_h^2 K_A - K_A^2/m - m r^4 h^4, \tag{9}$$

$$\mathcal{E}_m = B^{\mathrm{f}}_{22,m} + \tfrac{1}{4}\mathcal{L}_h^2 A_m. \tag{10}$$

Then $[h, D_{2,m}] = 0$ and

$$\begin{aligned} m Z_2 - D_{2,m} - \mathcal{E}_m = {} & (r/2 + r^2/4)\mathcal{L}_h^2 Y + 2r\mathcal{J}_h U + 3r^2 \mathcal{J}_h^2 Y \\ & - \big(\{K_A, Y\} + Y^2\big)/m. \end{aligned} \tag{11}$$

At fixed λ, the prior fixed-degree moment bounds give $r = O(p^{-1})$, $a_m = O(p)$ and bounded moments of Y, U, h on the central charge window. Thus the right-hand side is $O_{L^2}(p^{-1})$; the usual binomial tail absorbs its polynomial bounds off that window. This removes known return/drift contributions. It does not subtract the still unresolved conserved and quadratic-chaos pieces of $\mathcal{E}$.

3 A five-dimensional fourth-chaos problem

Let $\mathcal{D}X = Q_f X - \sum_i c_i^\dagger X c_i$ and $\mathcal{T}_k = (kp - \mathcal{D})H^k$. In the following definitions every operator is finally projected with the ordinary Gaussian-chaos map W_4. Introduce five Hermitian raw expressions:

$$E_0 = (AH^2 + H^2A)/2, \qquad M_0 = HAH, \tag{12}$$

$$S_0 = (H\mathcal{T}_3 + \mathcal{T}_3 H)/2 - E_0 - M_0, \tag{13}$$

$$C_0 = \mathcal{T}_4 - 2E_0 - M_0 - 2S_0, \qquad Z_0 = H^4. \tag{14}$$

Write $(E, M, S, C, Z) = W_4(E_0, M_0, S_0, C_0, Z_0)$. For four consecutive Hamiltonian slots these are respectively the single contraction marks

$$\tfrac{1}{2}(01 + 23), \qquad 12, \qquad \tfrac{1}{2}(02 + 13), \qquad 03, \qquad \text{no mark.} \tag{15}$$

They satisfy

$$W_4\mathcal{E} = C - 2S + \tfrac{1}{2}E + \tfrac{1}{2}M, \tag{16}$$

$$W_4(\mathcal{J}_H T_3) = S - \tfrac{1}{2}(E + M), \tag{17}$$

$$W_4(\mathcal{J}_H^2 A) = (E + M)/2, \qquad W_4(\mathcal{L}^2 A) = 2(E - M). \tag{18}$$

Thus E, M, S, Z span the fourth-chaos descendants of the already present derivative fields and the plain Hamiltonian polynomial. They are not all conserved operators.

The 24 pairings connecting two four-slot blocks determine their Gram matrix, with all internal block pairings removed by W_4. Project C onto the other four directions. With $d_3 = q^2 - q + 1$ and $d_5 = 1 + q + q^2 + q^3 + q^4$, the residual is

$$Q_4 = C - \alpha E - \beta M - \gamma S - \delta Z, \tag{19}$$

where

$$\alpha = -\frac{2q^3}{d_3 d_5}, \qquad \beta = -\frac{q^2(1 + q + q^2)}{d_3 d_5}, \tag{20}$$

$$\gamma = \frac{2q}{d_3}, \qquad \delta = \frac{2\lambda(1 - q)^3(1 + q)(1 + q + q^2)}{(1 + q^2)d_5}. \tag{21}$$

Consequently $W_4\mathcal{E}$ contains Q_4 with coefficient one, plus $(\alpha + 1/2)E + (\beta + 1/2)M + (\gamma - 2)S + \delta Z$. The projection coefficients depend on the limiting λ. They specify asymptotic Gram orthogonality, not exact finite-size orthogonality. The first-return remainder vanishes in this fourth chaos, and $W_4 D_{2,m} \to 0$, so this component is genuinely present in the second memory coefficient.

4 A complete endpoint certificate for the kernel

The variance $V_4 = \langle Q_4^2 \rangle_{\mathrm{DS}}$ is positive:

$$V_4 = \sigma^8 \frac{4\lambda(1-q)^2(1+q)^2\,[P(q) + 4\lambda q^3 T(q)]}{(1+q^2)d_3 d_5}, \tag{22}$$
$$P(q) = 1 + q + 2q^2 + q^3 + q^4, \qquad T(q) = 2 + q + 2q^2 + q^3 + 2q^4.$$

All factors in this expression are positive for $0 < q < 1$. The independently contracted dynamic curvature is

$$\frac{\langle Q_4 \mathcal{L}^2 Q_4 \rangle_{\mathrm{DS}}}{V_4} = 2\sigma^2(1 - q^4). \tag{23}$$

To obtain a full kernel rather than infer it from this moment, consider arbitrary $\langle Q_4 H^u Q_4 H^v \rangle$ with fixed u, v.

For each Gaussian pairing, the established local determinant rule for two single marks is $q^{\mathrm{cr}}(\lambda d + \lambda^2 ab)$, with

$$a_{ij} = 2 - 4\,\mathbf{1}_{\text{incident chords cross}}, \qquad d_{ij,kl} = 4\,\mathbf{1}_{\pi(i)=l,\ \pi(j)=k}. \tag{24}$$

Constants in (19) are included before summation. For a fixed word, Poisson overlap limits and exponential-moment bounds justify this rule, including fixed mark derivatives.

Group pairings by the number $k = 0, \ldots, 4$ of bridges between the two four-slot blocks. There are $8-2k$ remaining external endpoints, colored by which block they join, and an arc split between the blocks. Permuting slots inside the blocks produces an orbit of $24^2/k!$ distinct pairings. The full classification has

$$\sum_{k=0}^{4} \binom{8-2k}{4-k} (9-2k) = 807 \tag{25}$$

templates and 452760 pairing words. The recorded exact integer polynomial certificate gives zero for both the λ and λ^2 coefficients in every template with $k < 4$. The $k = 4$ coefficient agrees exactly with (22).

Spectator chords contribute a common factor to each orbit: moving an endpoint within a contiguous marked block cannot change its crossing with a chord whose endpoints are both external. Therefore this finite classification covers arbitrary spectator number; it is not a finite-order fit. The four surviving bridges cross each spectator four times. Hence all two-energy moments equal those of an existing probe with parameter $r = q^4$ and strength V_4.

Positive finite two-energy measures, higher even moment bounds and compact support of the candidate imply weak spectral convergence. In particular, for every fixed centre time T,

$$\langle Q_4^\dagger Q_4(T) \rangle_{\mathrm{DS}} = V_4 G_{q^4}(T). \tag{26}$$

The Mehler kernel itself is the established one [2].

5 Cross measures and the small-coupling channel

A second endpoint calculation pairs a four-slot Q_4 block with the two-slot Y or three-slot R block. There are respectively 128 and 354 templates. Each is zero as a polynomial in q at both powers of λ. The same spectator argument covers all fixed two-energy moments. Spectral Cauchy–Schwarz, tightness and compact moment determinacy therefore give

$$d\nu_{Q_4,Y} = 0, \qquad d\nu_{Q_4,R} = 0. \tag{27}$$

This is energy-resolved two-point orthogonality, so it also holds after multiplying either lower field by fixed left/right energy polynomials. No mixed higher-point independence has been assumed.

Since $0 < q^4 < 1$, the limiting Mehler measure is absolutely continuous and has no diagonal atom. The exact zero-frequency component of Q_4 vanishes in the limit, by the closed-diagonal Portmanteau bound. The field is a dynamic channel rather than a residual conserved polynomial. This fact does not fix its bulk mass.

The subsequent small-λ limit is

$$V_4 \sim \frac{768}{5}\frac{J_s^8}{\lambda}, \qquad \frac{\langle Q_4^\dagger Q_4(T)\rangle}{V_4} \longrightarrow \operatorname{sech}^8(2J_sT), \qquad \mu_2 \longrightarrow 32J_s^2. \tag{28}$$

Here $\Delta = 4$ is a chord probe label. The single-time kernel and its frequency spectrum must still be matched to actual spatial insertions before a meson interpretation can be completed.

For $J_s = 1$, independent evaluations of the normalized kernel give

λ	$T = 0.25$	$T = 0.5$	$T = 1$
0.25	0.7799243687	0.3547099044	0.0044426013
0.10	0.6052107914	0.1290475638	0.0002415544
0.025	0.4458017205	0.0474957676	0.0000461996
0.01	0.4082285863	0.0370478710	0.0000320903

The finite-coupling dependence is substantial. The Jacobi formula is

$$G_{q^4}(T) = \sum_n q^{4n} |\langle n|e^{-iTT_q}|0\rangle|^2, \qquad (T_q)_{n-1,n} = \sigma\sqrt{\frac{1-q^n}{1-q}}. \tag{29}$$

Its vacuum Taylor truncation bound is below 10^{-13}, with a size increase of 12. At $\lambda = 0.25, 0.1$ the result is also checked against 128/192-node Mehler quadrature. This separates truncation control from floating-point refinement and from the physical small-λ approximation.

6 What remains in the raw fourth coefficient

The second memory coefficient is quartic in the Gaussian couplings. After subtracting its ensemble scalar part, it still has both second and fourth coupling chaos. The present result resolves a new fourth-chaos channel and its cross measures with Y, R; it does not identify $\mathcal{E}$ or mZ_2 with Q_4 alone.

The immediate continuation is to compute $W_2\mathcal{E}$, its conserved H^2 projection and any rank-filtered dynamic remainder. One must also translate the fourth-chaos descendants E, M, S, Z back into full operators with their lower-chaos contractions restored. Only then can the actual second memory coefficient have a complete two-point spectral decomposition. Mixed correlations with F and composite fields such as Y^2 require their own same-J analysis.

For finite memory parameters, the two-insertion remainder still needs a bound at the collective normalization. The first three Taylor coefficients, even when individually solved, do not establish that expansion's convergence after double scaling. The next field therefore supplies concrete spectral data for reconstruction, while the finite-memory and bulk spatial problems retain their full scope.

Method and literature boundary. Interval-partition/first-return algebra is standard [1]. The endpoint cancellation argument identifies a particular microscopic composite with an existing chord measure; it is not a new derivation of the Mehler formula. We also examined the composite-operator relations for analytically continued degenerate matter in [3]. Their negative-dimension insertion is a distinct object; its fusion rules do not by themselves identify the present positive fourth-chaos field. No such identification is used here.

References

[1] O. Arizmendi, M. Banna and P.-L. Tseng, *Quantitative Estimates for Operator-Valued and Infinitesimal Boolean and Monotone Limit Theorems*, arXiv:2211.08054, Sec. 2.1, Definition 2.4.

[2] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2.

[3] K. Hosomichi and M. Sato, *Degenerate chords in double-scaled SYK*, arXiv:2512.01177v2, Sec. 4, especially Eqs. (4.2)–(4.7).